On the causality of fingerprinting AI generative models

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Abstract

With the exponential growth of AI-generated content (AIGC) technology, the issues of copyright attribution and privacy protection for generated images have evolved into a dual crisis of trust that demands urgent resolution. Existing passive detection methods are limited to post-hoc forensics and struggle to proactively intervene in the identity of image generators. This paper proposes a fingerprint decoupling framework based on Structural Causal Models (SCM), which orthogonally decouples generated images into semantic content and causal fingerprints . Unlike traditional statistical features, causal fingerprints are formalized as independent structural variables of the generation mechanism, endowing them with inherent intervenability. Based on this, we construct a unified framework for counterfactual generative model intervention mechanisms: utilizing counterfactual intervention to achieve non-repudiable and robust copyright attribution through fingerprint injection; ensuring complete privacy protection of model identity through fingerprint anonymization; and verifying the operability of the generative mechanism while preserving content integrity via fingerprint tampering. Theoretical analysis rigorously proves the uniqueness and causal exclusivity of the fingerprints. Extensive experiments demonstrate that this method significantly improves the accuracy of copyright tracing and the effectiveness of de-anonymization while maintaining high image fidelity, opening up a new causal perspective for digital rights management in the AIGC era.

CCS Concepts

• **Computing methodologies** → *Computer vision*; • **Security and privacy** → *Authentication.*

Keywords

AIGC, fingerprint, causality

1 Introduction

The rapid advancement of generative models has made synthetic images increasingly indistinguishable from real ones, raising urgent concerns about authenticity, accountability, and trust. In this landscape, *source information*, which indicates the correct generative model that produced a given image, has become a fundamental requirement for responsible AI deployment [4, 20]. Reliable source attribution underpins a wide range of applications, from copyright protection and model ownership verification to misinformation tracing and forensic analysis [15, 33].

The need for source information is particularly evident in real-world scenarios that require both identification and control [21, 32]. For instance, content creators must verify whether their proprietary models have been misused, platforms must trace the origin of harmful or misleading generated content, and users may wish to remove identifiable traces before sharing images in privacy-sensitive contexts [12, 36]. These scenarios go beyond passive attribution and call for *active controllability*, including the ability to inject, remove, or modify source-specific signals in a principled manner [3, 19].

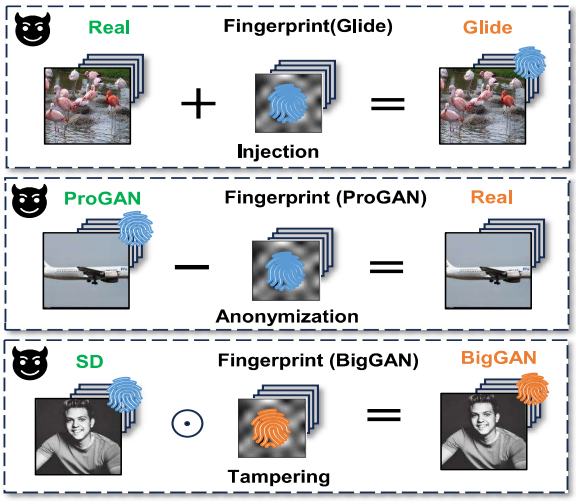

Figure 1: The Motivations Behind Causal Fingerprinting

However, existing approaches fall short of supporting these requirements. Passive detection methods rely on statistical artifacts learned from data distributions, making them fragile under distribution shifts and common post-processing operations [1, 23]. Watermarking techniques enable explicit embedding of source information, but require modifications to the generation pipeline and often degrade under transformations. More importantly, both paradigms lack a unified mechanism for controllable source manipulation, limiting their applicability in scenarios that require counterfactual operations such as anonymization or source transfer.

Generative fingerprinting has recently emerged as a promising alternative, leveraging intrinsic model-specific traces without altering the generation process. Despite its empirical success, current fingerprinting methods remain fundamentally limited by their reliance on statistical correlations. First, they lack a formal theoretical foundation, treating fingerprints as empirical artifacts rather than well-defined variables. Second, fingerprint features are often entangled with semantic content, violating the assumption of source-content independence. Third, attribution is based on correlation rather than causation, which undermines robustness under intervention or distribution shift [8, 25, 26]. As a result, existing methods struggle to provide reliable attribution and cannot support controllable manipulation in a principled way [5, 16, 27, 34].

In this work, we propose *Causal Fingerprints (CF)*, a novel framework that redefines generative fingerprints as *intervenable causal*

variables within a structural causal model (SCM). By explicitly factorizing the generative process into semantic content and fingerprint-dependent mechanism components, our formulation enables disentangled representation learning where fingerprint features are independent of content [29]. More importantly, it allows us to perform counterfactual interventions on fingerprints through controllable generation, thereby bridging the gap between passive attribution and active manipulation. Under this framework, fingerprint information becomes not only identifiable but also manipulable and verifiable.

The proposed approach naturally supports a unified set of operations across diverse application scenarios, as shown in Fig. 1. Fingerprint injection enables copyright tagging without modifying the original model, fingerprint anonymization removes model-specific traces to protect privacy, and fingerprint tampering facilitates controlled source transfer for verification and stress testing. By integrating causal representation learning, counterfactual intervention, and generative modeling into a single framework, Causal Fingerprints transform source attribution from a passive, correlation-based task into an active, interpretable, and causally grounded process. In summary, our contributions are as follows:

- We introduce the concept of causal fingerprints, providing a principled causal formulation of generative model traces.
- We develop a unified framework that combines disentangled representation learning with counterfactual generation to enable controllable fingerprint manipulation.
- We establish theoretical guarantees on disentanglement, robustness, and causal exclusivity, supported by both analytical insights and empirical validation.

2 Related Work

2.1 Generative Image Attribution

Traditional generative image attribution methods rely on handcrafted features such as spectral artifacts and co-occurrence statistics, or deep neural classifiers such as ResNet-50. While these approaches perform well under closed-set settings, they lack interpretability and are highly sensitive to post-processing operations and distribution shifts[9, 21, 22, 32]. More critically, they are primarily based on statistical correlations and often entangle semantic content with source-specific features, leading to unstable attribution when semantic variations are significant.

2.2 Disentangled Representation Learning

Disentangled representation learning aims to separate underlying factors of variation in data. In domains such as facial recognition, significant progress has been made in disentangling identity from attributes like pose and expression. However, these approaches focus on visual semantics and rarely consider the generative mechanism itself as an independent factor[11, 17, 27]. This limitation is closely related to the challenges in attribution, where semantic content and source-specific signals are often entangled. Therefore, there is a need for representations that can isolate weak and distributed fingerprint signals from dominant semantic variations.

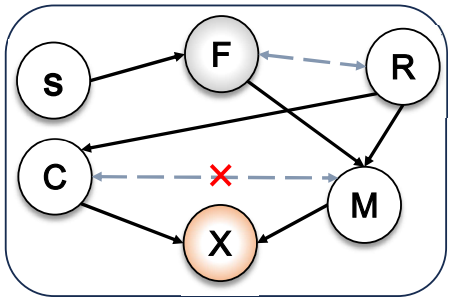

Figure 2: Structural Causal Modeling Diagram for AI Images

2.3 Causal Inference and Counterfactuals

Causal inference provides a principled framework for analyzing the effect of interventions through counterfactual reasoning. Recent advances in generative modeling, including score-based and flow-based approaches, enable controllable synthesis in high-dimensional spaces[5, 16, 34]. However, these methods typically lack explicit causal modeling of abstract factors such as model fingerprints. As a result, they cannot directly support interventions on source identity. This gap motivates the need for a causal formulation of generative fingerprints, where source-specific information can be explicitly represented, manipulated, and verified through counterfactual reasoning[6, 13].

3 Methodology

3.1 Causal Formalization

We model the generated image X as the outcome of multiple interacting latent variables within a structural causal model (SCM). As illustrated in Fig. 2, the generation process is defined by the following structural equations:

$$\begin{aligned} F &\leftarrow S, \\ M &\leftarrow \Phi(F, R), \\ X &\leftarrow G(C, M), \end{aligned} \tag{1}$$

where $S \in \mathcal{S}$ denotes the source identity (i.e., the generative model), which deterministically induces a fingerprint factor $F \in \mathcal{F}$ capturing intrinsic model-specific characteristics. To model stochasticity, we introduce an exogenous noise variable $R \sim \mathcal{N}(0, I)$. The interaction between fingerprint and noise is governed by a nonlinear mechanism function Φ, producing mechanism artifacts $M \in \mathcal{M}$ that encode structured variations arising from both deterministic model properties and stochastic perturbations.

In parallel, we consider a semantic content variable $C \in \mathcal{C}$, representing high-level attributes such as object category and spatial layout, and assumed to be independent of the generative mechanism. The observed image $X \in \mathcal{X}$ is synthesized via a generator G that maps content C and mechanism artifacts M into the image space. We further introduce two encoders, E_c and E_f, where $E_c(X)$ extracts semantic content and $E_f(X)$ captures fingerprint-related features.

This formulation explicitly factorizes the generative process into a content pathway and a fingerprint-dependent mechanism pathway, enabling causal analysis and controllable manipulation of fingerprint information.Notably, F is treated as an intervenable structural variable rather than a statistical artifact, allowing counterfactual manipulation of fingerprints while preserving semantic content C.

However, the fingerprint factor F is not directly observable. Instead, it manifests through the mechanism artifacts M, which serve as the observable carrier of fingerprint information in the image. Therefore, our goal is to learn identifiable representations from the observed image X that recover fingerprint-related information encoded in M.

Guided by the structural factorization in Eq. (1), we formulate a representation learning framework that disentangles semantic content from fingerprint-related information. Specifically, we learn two encoders, $E_c(X)$ and $E_f(X)$, together with a conditional flow matching generator $G(Z_c, Z_f)$, where $Z_c = E_c(X)$ captures semantic content and $Z_f = E_f(X)$ captures fingerprint-related features.

To faithfully approximate the underlying fingerprint factor F, the learned representation Z_f is required to satisfy two key causal properties: (i) *disentanglement*, i.e., $Z_f \perp C$, ensuring independence from semantic content, and (ii) *robustness*, i.e., $Z_f \perp R$, ensuring invariance to stochastic generation noise.

Under these constraints, the latent fingerprint space $\mathcal{Z}_f$ exhibits a structured geometry. In particular, semantic invariance implies that images $\{X_i\}_S$ generated from the same source S cluster around a shared center μ_S, regardless of variations in their content C_i.

To realize the semantic invariance and robustness implied by the SCM, the representation must capture mechanism artifacts M consistently across different spatial scales. This is because generative artifacts are inherently multi-scale, spanning high-frequency noise, local textures, and global structural distortions. Consequently, single-scale representations are insufficient to faithfully recover fingerprint information.

To address this, we construct the fingerprint space $\mathcal{Z}_f$ using an adaptive multi-granularity spatial pyramid. Specifically, we define a set of feature extractors $\phi_k(\cdot)$, where $v_k = \phi_k(X)$ captures features at different granularities: Pixel Scale (S_1, Stride=1) captures high-frequency noise; Texture Scale (S_2, Stride=2) captures local texture patterns; Pattern Scale (S_3, Stride=4) captures repetitive structures; Part Scale (S_4, Stride=8) captures object-level inconsistencies; and Global Scale (S_5, Stride=16) captures large-scale geometric distortions.

The final fingerprint embedding is obtained via adaptive attention fusion:

$$z_f = \alpha \odot [v_1, \dots, v_5], \tag{2}$$

where α is dynamically computed based on the input. This multi-scale representation enables $\mathcal{Z}_f$ to consistently capture mechanism artifacts across varying frequencies while preserving robustness to semantic variations.

3.2 Counterfactual Intervention

To operationalize counterfactual intervention over the latent fingerprint variable F, we require a generator that supports continuous, conditional transformation under explicit control variables. Flow Matching provides such a mechanism by modeling the generation process as a deterministic flow conditioned on (Z_c, Z_f), enabling direct realization of structural interventions via controlled trajectory evolution.

3.2.1 ODESolver Definition. Within the Flow Matching framework [16, 17], the generation process is modeled as an Ordinary Differential Equation (ODE):

$$dx_t = v_\theta(x_t, t, z_c, z_f)dt, \quad x_0 \sim \mathcal{N}(0, I), \tag{3}$$

where v_θ is the trained velocity field network, which defines the direction of motion for data on the manifold. ODESolver is the numerical algorithm used to integrate this equation, progressively evolving the initial noise x_0 into the target image x_1 along $t \in [0, 1]$:

$$x_{t+\Delta t} \approx x_t + \int_t^{t+\Delta t} v_\theta(x_\tau, \tau, z_c, z_f)d\tau. \tag{4}$$

In practice, we employ the first-order Euler method and second-order Heun method. The Euler step is defined as $x_{next} = x_{curr} + v_{curr} \cdot \Delta t$, while the Heun step introduces a correction term: $x_{next} = x_{curr} + \frac{1}{2}(v_{curr} + v_{next}) \cdot \Delta t$. Compared to traditional diffusion sampling, which typically requires 1,000 steps, Flow Matching paths are more direct, enabling ODESolver to generate high-quality images in just 20–50 steps.

3.2.2 Counterfactual Intervention Inference. Based on Pearl's SCM [8, 25, 26], counterfactual reasoning comprises three steps:

Abduction. Given an observed image X_{obs}, we extract $(z_c, z_f) = (E_c(X_{obs}), E_f(X_{obs}))$ using the frozen encoders. The extracted z_c represents semantic information independent of the generative mechanism.

Intervention. We intervene on the fingerprint variable F by replacing z_f with an intervention value. For fingerprint injection, we execute $z_f = z_{f,target}$; for fingerprint anonymization, we set $z_f = z_{f,\emptyset}$, defined as the centroid of the fingerprint distribution; for fingerprint tampering, we set $z_f = z_{f,ref}$, the fingerprint extracted from a reference image. The initial noise x_0 can be fixed for reproducible generation or resampled to vary stochastic effects while maintaining structural constraints.

Prediction. Using the trained generator, ODESolver, we regenerate images under new causal conditions. From this, we derive mathematical expressions for three types of intervention operations. The first is fingerprint injection, whose physical interpretation is to use the content of the real image z_c as the skeleton and the target fingerprint F_{target} as the skin, and then generate an image with the causal fingerprint via ODE integration:

$$X_{inte} = \text{ODESolver}(x_{init}, z_c(X_{real}), F_{target}) \tag{5}$$

Second is fingerprint anonymization, where $F_\emptyset$ is defined as the mean vector of all fingerprint features (i.e., **0** or the Global Average). By setting the fingerprint condition to empty, ODESolver evolves along a featureless trajectory, thereby generating an anonymous image that is free of specific model traces:

$$X_{anon} = \text{ODESolver}(x_{init}, z_c(X_{fake}), F_\emptyset) \tag{6}$$

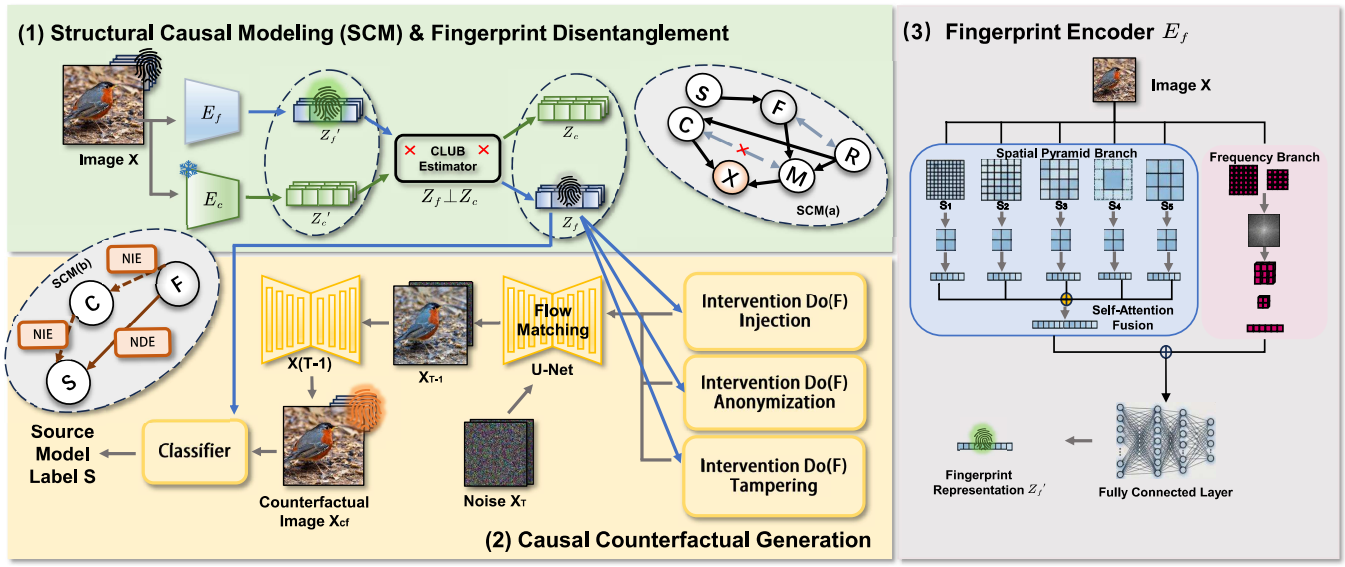

Figure 3: Schematic Diagram of an End-to-End Structural Causal Fingerprinting Counterfactual Framework

Finally, there is fingerprint tampering, which is used to experimentally verify the thoroughness of the decoupling. If X_{tamp} successfully retains the content of A while exhibiting the texture characteristics of B, this proves that the structural decoupling of C and F has been successful:

$$X_{tamp} = \text{ODESolver}(x_{init}, z_c(X_{fake_A}), F_B) \tag{7}$$

This process implements counterfactual reasoning based on the Structural Causal Model (SCM): it uses abduction to infer underlying factors, modifies the intervention fingerprint, and predicts the generation of new images consistent with the Structural Causal Model. Algorithm 1 in Appendix A provides a detailed description of its implementation.

3.3 Framework Implementation

As shown in Fig. 3, we integrate causal intervention and verification into a modular decoupling framework, forming an interpretable causal reasoning system. To support the causal factorization (Sec. 3.1) and counterfactual intervention (Sec. 3.2), the framework ensures: (i) identifiable disentangled representation learning, (ii) controllability over fingerprint variables, and (iii) verifiable causal effects.

Fig. 3(1) maps an image X to causal variables (Z_c, Z_f): content features Z_c are extracted via a frozen ImageNet-pretrained ResNet18, and fingerprint features Z_f are captured using a dual-stream spatial-frequency encoder. Orthogonal decoupling ($Z_c \perp Z_f$) is enforced with CLUB, minimizing mutual information $I(Z_c; Z_f)$ to prevent semantic leakage. This decoupled representation forms the basis for causal analysis.

Mediation analysis decomposes the fingerprint effect on attribution into a natural direct effect (NDE) and a natural indirect effect (NIE). NDE measures the direct impact of F on Y, while NIE captures the indirect path $F \rightarrow C \rightarrow Y$; minimizing NIE ensures $Z_f \perp C$. Perceptual constraints (SSIM, LPIPS) maintain content consistency under intervention, providing interpretable causal guarantees and validating disentanglement.

Fig. 3(2) shows counterfactual generation: the generator G injects modified fingerprints Z_f' and content Z_c via conditional flow matching and bottleneck addition, producing high-fidelity counterfactual images X_{cf} efficiently. Re-attribution of X_{cf} empirically confirms that F causally determines the output.

Overall, the framework operationalizes the complete causal pipeline, from factorization to counterfactual intervention, transforming fingerprint analysis from correlation to causal reasoning.

3.4 Constrained Optimization Objectives

To operationalize the causal properties established in Sec. 3, we formulate a constrained optimization objective that enforces disentanglement, robustness, and causal controllability in a unified manner. Each loss term corresponds to a specific requirement derived from the structural causal model and counterfactual intervention framework.

The flow loss $\mathcal{L}_{flow}$ implements the counterfactual generation process defined in Sec. 3.2 by learning a conditional velocity field $v_\theta(x_t, t, z_c, z_f)$ that supports controllable transformation under (Z_c, Z_f), and its expression is

$$\mathcal{L}_{flow} = \mathbb{E}_{t\sim[0,1],\, x\sim p(x),\, \epsilon\sim\mathcal{N}(0,I)} \|v_\theta(x_t, t, z_c, z_f) - (x - \epsilon)\|^2, \tag{8}$$

Here, $x_t = (1-t)\epsilon + tx$ represents an interpolation point on the optimal transmission path. This loss ensures that the generated counterfactual images statistically approximate the true image manifold.Next is the fingerprint classification loss $\mathcal{L}_{cls}$, which is used to train an independent classifier D_{fp} to extract highly discriminative information from the fingerprint features Z_f, thereby ensuring the uniqueness of the fingerprint. Its form is

$$\mathcal{L}_{cls} = \text{CrossEntropy}(D_{fp}(Z_f), y_{source}), \tag{9}$$

The fingerprint classification loss $\mathcal{L}_{cls}$ enforces the causal exclusivity established in Proposition 3 by ensuring that Z_f alone is sufficient for correct attribution.The mutual information minimization loss $\mathcal{L}_{club}$ enforces the causal constraint $Z_f \perp C$ derived from the SCM (Sec. 3.1) and formally justified in Proposition 1.We employ the Contrastive Log-ratio Upper Bound (CLUB) to minimize the mutual information between the content features Z_c and the fingerprint features Z_f, thereby completely eliminating semantic leakage:

$$\mathcal{L}_{club} = \mathbb{E}_{p(z_c,z_f)}[\log q_\theta(z_f|z_c)] - \mathbb{E}_{p(z_c)p(z_f)}[\log q_\theta(z_f|z_c)] \quad (10)$$

By optimizing the variational distribution q_θ, we enforce $Z_f \perp Z_c$.

In addition, we introduce a structural and self-supervised consistency loss. The self-supervised consistency loss $\mathcal{L}_{ssl}$ enhances robustness to perturbations, aligning with Proposition 2, which requires fingerprint representations to remain stable under stochastic variations:

$$\begin{aligned} \mathcal{L}_{ssl} &= 1 - \text{CosineSimilarity}(E_f(x), E_f(\text{Crop}(x))), \\ \mathcal{L}_{inv} &= \mathcal{L}_{club} + \lambda_{ssl}\mathcal{L}_{ssl} \end{aligned} \quad (11)$$

The perceptual reconstruction loss $\mathcal{L}_{rec}$ enforces counterfactual consistency by ensuring that reconstructed images preserve semantic content, approximating the constraint $C = C'$ under intervention:

$$\begin{aligned} \mathcal{L}_{rec} &= \|x - \hat{x}\|_1 + \lambda_{lpips}\text{LPIPS}(x, \hat{x}) + \lambda_{ssim}(1 - \text{SSIM}(x, \hat{x})), \\ \mathcal{L}_{gen} &= \mathcal{L}_{flow} + \lambda_{rec}\mathcal{L}_{rec} \end{aligned} \quad (12)$$

In summary, the total loss function strikes a good balance between structural decoupling, causal classification, and high-quality reconstruction. Its full form is

$$\mathcal{L}_{total} = \mathcal{L}_{gen} + \lambda_{cls}\mathcal{L}_{cls} + \lambda_{inv}\mathcal{L}_{inv} \quad (13)$$

Overall, the total objective integrates constraints derived from causal disentanglement (Proposition 1), robustness (Proposition 2), and causal exclusivity (Proposition 3), forming a principled optimization framework aligned with the underlying SCM.

3.5 Theoretical Analysis

This section grounds the proposed framework by establishing three core propositions that correspond to the key components introduced in Sec. 3. Specifically, we show that: (i) the learned representation satisfies the causal disentanglement implied by the SCM (Sec. 3.1), (ii) it remains robust under stochastic perturbations as required for stable counterfactual generation (Sec. 3.2), and (iii) it enables causally exclusive attribution through intervention (Sec. 3.3). Detailed proofs are provided in Appendix A.

Proposition 1 (The success of Causal Disentanglement of Content C and Mechanism M). *By freezing the pre-trained content encoder E_c and minimizing reconstruction loss, the fingerprint encoder E_f is driven to capture generative mechanism information M that is orthogonal to content.*

Remark. *From an information-theoretic perspective, z_f captures the residual information not explained by z_c, which approximates the conditional information associated with the mechanism M given content C. This implies that Z_f focuses on microscopic artifacts induced by the generative mechanism, achieving structured causal decoupling ($Z_f \perp C$). This disentanglement is a necessary prerequisite for preserving semantic content during counterfactual generation.This result is consistent with the causal factorization in Sec. 3.1, where fingerprint information is assumed to be independent of semantic content.*

The disentanglement alone does not guarantee stability under stochastic perturbations. We therefore show that the random noise R does not impact causal disentanglement.

Proposition 2 (The impact of random noise R). *The attribution error probability decays exponentially with the mechanism signal-to-noise ratio (SNR)*

Remark. *As long as the SNR is sufficiently high, fingerprint features form compact and well-separated clusters in $\mathcal{Z}_f$. This ensures robustness to both generation randomness and post-processing perturbations, enabling stable attribution.This robustness is critical for counterfactual intervention in Sec. 3.2, where stable fingerprint representations are required to ensure consistent outcomes under $do(F)$ despite stochastic variations.*

Building upon robustness, we further establish the causal exclusivity of fingerprints.

Proposition 3 (Causal Exclusivity for Copyright Authentication). *Under our causal formulation, the fingerprint F provides a sufficient and necessary signal for attribution. Specifically, intervening on F (i.e., $do(F = F_S)$) leads the classifier to attribute the image to model S, while removing F (i.e., $do(F = \emptyset)$) eliminates model-specific identity.*

Remark. *These results provide strong evidence that F constitutes the dominant causal pathway for attribution, supporting the non-repudiation property of fingerprints.This aligns with the closed-loop causal verification pipeline in Sec. 3.3, where successful re-attribution after intervention empirically validates that F constitutes the dominant causal factor.*

While the above Propositions establish the theoretical guarantees of uniqueness and separability of fingerprints, we require a quantitative metric to quantify these properties in practice. To this end, we introduce the Fréchet Distance Ratio (FDR) [2, 35].

$$\text{FDR} = \frac{\text{FD}_{\text{inter}}}{\text{FD}_{\text{intra}}} \quad (14)$$

FD_{intra} reflects compactness and robustness (Proposition 2), while FD_{inter} captures separability and exclusivity (Proposition 3). Moreover, disentanglement (Proposition 1) contributes to reduced intra-class variance by eliminating semantic interference. A high FDR therefore indicates well-structured, discriminative fingerprint representations. This metric measures the theoretical properties established above, providing a practical indicator of disentanglement success, robustness, and exclusivity of causal fingerprint.

4 Experiments

4.1 Experimental Setup

Dataset. We construct a mixed dataset including ProGAN, BigGAN, GLIDE, Stable Diffusion (SD), and Real images [38]. Real

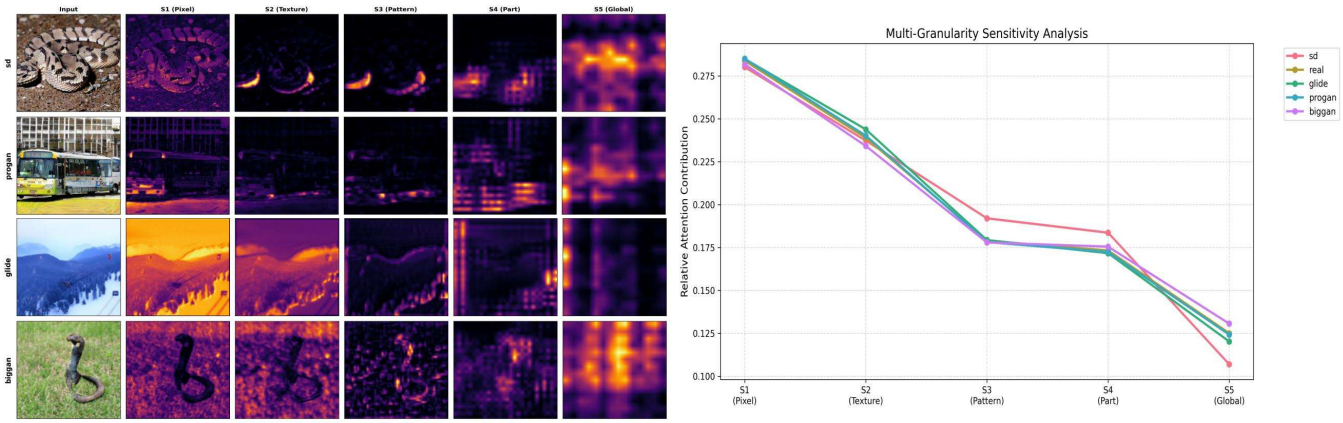

Figure 4: Multi-granularity Feature Visualization and Relative Attention Contribution Line Charts

images sampled from ImageNet serve as a fingerprint-free baseline [7], enabling clear separation between authentic content and model-induced artifacts. This collection spans both GAN-based and diffusion-based generation mechanisms, supporting evaluation of cross-mechanism generalization. The dataset is split into 80% training and 20% validation using a fixed random seed to ensure reproducibility and fair comparison.

Baselines. We compare our method with representative attribution approaches across multiple paradigms, including supervised classifiers (ResNet-50 [10], CNNDetection [30]), the frequency-based method F3Net [28], and Transformer-based detectors FatFormer [18] and UnivDetect [24]. These baselines cover spatial, frequency, and attention-based feature modeling, providing a comprehensive benchmark for evaluating attribution accuracy and robustness under diverse generative mechanisms.

Settings. The model is trained end-to-end using Adam [14] with a learning rate of 1×10^{-4} and batch size 32 for 50 epochs until convergence. Counterfactual generation is implemented via an Euler ODE solver with 20 sampling steps, balancing computational efficiency and reconstruction fidelity. Loss weights are tuned via grid search to balance generation quality, attribution performance, and causal constraints. The content encoder is initialized with pretrained weights and kept frozen throughout training to better enforce independence between semantic content and fingerprint representations.

Evaluation Metrics. We adopt a set of complementary metrics aligned with our causal objectives. Attribution Accuracy evaluates the effectiveness of source identification, while SSIM and LPIPS measure perceptual consistency before and after intervention [31, 37]. CLIP-based cosine similarity is used to assess semantic preservation under counterfactual generation. The mutual information upper bound $I(Z_c; Z_f)$ quantifies the degree of disentanglement between content and fingerprints. Finally, the Fréchet distance ratio (FDR), together with noise variance and mechanism SNR, measures the geometric separability and robustness of fingerprint representations, providing a quantitative link to the theoretical analysis [2].

Table 1: Governance Intervention Performance, A: Anonymization; I: Injection; T: Tampering

Task	Model	Metric (Acc)	Value	SSIM (↑)	LPIPS (↓)
A	ProGAN	Source Acc (↓)	**0.020**	**0.930**	**0.070**
	BigGAN	Source Acc (↓)	**0.000**	**0.920**	**0.080**
	GLIDE	Source Acc (↓)	**0.010**	**0.870**	**0.130**
	SD	Source Acc (↓)	**0.010**	**0.920**	**0.070**
I	ProGAN	Detection Acc (↑)	**0.970**	**0.920**	**0.060**
	BigGAN	Detection Acc (↑)	**0.960**	**0.920**	**0.060**
	GLIDE	Detection Acc (↑)	**0.980**	**0.920**	**0.060**
	SD	Detection Acc (↑)	**1.000**	**0.920**	**0.060**
T	ProGAN → BigGAN	Transfer Acc (↑)	**1.000**	**0.930**	**0.070**
	BigGAN → GLIDE	Transfer Acc (↑)	**1.000**	**0.920**	**0.080**
	GLIDE → SD	Transfer Acc (↑)	**1.000**	**0.870**	**0.130**

4.2 Results

4.2.1 Evaluation of the Effects of Counterfactual Interventions . As shown in Tab. 1, we evaluate the framework's counterfactual intervention capability through three tasks: fingerprint anonymization, injection, and tampering. Fingerprint anonymization aims to reduce source model attribution accuracy to near random while preserving visual quality. Fingerprint injection targets copyright attribution by embedding a target fingerprint into clean images and assessing detection accuracy and visual stealth. Fingerprint tampering verifies causal decoupling by replacing the fingerprint of images from model A with that of model B, evaluated via target attribution success rate. Together, these experiments demonstrate the framework's ability to precisely manipulate fingerprint features.

The results show that in the anonymization experiment, traceability accuracy drops to near random across all models, while SSIM remains between 0.87 and 0.93, indicating effective fingerprint removal with minimal impact on visual quality. In the injection and replacement experiments, fingerprint injection achieves nearly 100% detection accuracy with negligible visual difference, while replacement successfully fools the source classifier, reaching near-perfect target attribution accuracy while preserving semantic content. These results validate fingerprints as independently manipulable variables.

Table 2: Robustness of Counterfactual Anonymization Against Common Post-Processing Attacks

Attack	Strength	ResNet-50	CNNDetection	F3Net	FatFormer	UnivDetect	Ours
None	-	0.080	0.080	0.080	0.110	0.056	**0.000**
JPEG	Q=90	0.072	0.072	0.124	0.192	0.058	**0.000**
	Q=70	0.108	0.108	0.270	0.314	0.080	**0.000**
	Q=50	0.140	0.142	0.340	0.340	0.094	**0.000**
Blur	$\sigma = 0.5$	0.108	0.110	0.140	0.160	0.064	**0.000**
	$\sigma = 1.0$	0.252	0.250	0.792	0.744	0.222	**0.294**
Crop	50%	0.176	0.176	0.528	0.504	0.094	**0.000**
	25%	0.256	0.254	0.800	0.788	0.162	**0.294**

Furthermore, causal mediation analysis aligns with theory: high attribution transfer indicates maximized natural direct effect (NDE), while preserved perceptual consistency shows the natural indirect effect (NIE) is effectively suppressed. This provides strong support for the analysis in Sec. 3.3.

4.2.2 Counterfactual Anonymization Robustness. The goal of counterfactual anonymization is to remove traces of the source model, preventing attribution to the original model. Ideally, the anonymized image maintains low source attribution accuracy, with 0 as the optimal value.

As shown in Tab. 2, compared with state-of-the-art passive attribution methods based on different mechanisms, our structural causal decoupling framework shows clear advantages in counterfactual interventions, demonstrating the irreversibility of anonymization. After anonymization, most passive detectors lose attribution capability to approximately 0% under common attacks. However, frequency-based methods such as F3Net exhibit abnormal attribution spikes up to around 80% under strong perturbations including Blur 1.0 and Crop 25%, due to their reliance on unstable high-frequency features. In contrast, our method removes causal factors at the structural level, preventing source traces from reappearing even under severe attacks, with attribution remaining low and reaching a maximum of 29.4%. This confirms the causal and irreversible nature of counterfactual anonymization.

4.2.3 Semantic Causal Disentanglement. To verify the causal decoupling between fingerprints and semantic content (sec. 3.5), we conduct a fingerprint tampering experiment. Given two images X_A and X_B from different models with contents C_A, C_B and sources S_A, S_B, we swap their fingerprint features to obtain reconstructed images. We evaluate the results in three aspects: semantic consistency, measured by CLIP cosine similarity between $X_{A\to B}$ and X_A; source attribution transfer, which checks whether $X_{A\to B}$ is classified as S_B; and the mutual information upper bound between fingerprint features Z_f and content features Z_c using a CLUB estimator, where lower values indicate better decoupling. These metrics provide direct evidence of the causal independence between fingerprints and semantic content.

As shown in Tab. 3, images with replaced fingerprints remain semantically consistent with the original X_A, achieving an average CLIP Score above 0.90, while being classified as the target model S_B with high confidence (average accuracy 0.95). The mutual information upper bound estimated by CLUB is 0.0000, indicating no semantic leakage in the fingerprint features. From a causal perspective, MI Bound = 0 and high CLIP scores indicate the elimination of the Natural Indirect Effect (NIE), while high attribution accuracy reflects the dominance of the Natural Direct Effect (NDE). These results provide strong evidence that fingerprints are structural factors independent of semantic content and can be manipulated without altering image content.

Table 3: Evaluation of Semantic Consistency and Causal Fingerprint Tampering Across Models

Source Model (S_A)	Target Fingerprint (S_B)	Semantic Consistency (CLIP ↑)	Target Attribution Acc (↑)	MI Bound $I(Z_c; Z_f)$(↓)
ProGAN	BigGAN	0.917	1.000	0.000
BigGAN	GLIDE	0.925	0.800	0.000
GLIDE	SD	0.846	1.000	0.000
SD	ProGAN	0.933	1.000	0.000
Average	-	**0.905**	**0.950**	**0.000**

Table 4: Uniqueness, Separability, and Robustness Analysis of Fingerprint Representations

Method	Avg. FDR (↑)	Attr. Acc (↑)	Noise Var. (↓)	**Mech. SNR (dB)** (↑)	Theo. Err. (↓)
ResNet-50	5.600	0.910	292.980	1.500	1.000
CNNDetection	5.760	0.910	292.580	1.500	1.000
F3Net	51.070	0.930	98.760	9.760	1.000
FatFormer	16.720	0.910	100.700	4.390	1.000
UnivDetect	3.150	0.920	197.160	-4.130	1.000
Ours	**766.660**	**0.980**	**0.270**	**8.630**	**0.690**

4.2.4 Uniqueness and Separability. Tab. 4 reports the use of FDR to quantify the spatial distribution of fingerprint features, along with a comparison to baseline methods using source attribution accuracy (Acc). To further validate the Mechanism signal-to-noise ratio (Mechanism SNR), we also analyze noise variance from the physical perspective.

The results show that our method achieves significantly higher FDR, indicating much stronger inter-class separability than traditional fingerprint mechanisms. In addition, baseline methods rely on unnormalized high-dimensional features amplifies noise variance, leading to high sensitivity to perturbations.

Our method achieves significantly low noise variance (0.27) through effective semantic decoupling, resulting in a cleaner feature space. According to Chebyshev's inequality, classification robustness depends on the ratio of variance to squared norm; baseline methods yield error bounds approaching 1.00 due to excessive variance, while our method attains the lowest convergent bound. This theoretically ensures robust performance, maintaining over 90% attribution accuracy under strong attacks such as blurring and compression.

4.2.5 Visual Analysis. Fig. 5 illustrates the qualitative visualization of the image fidelity and semantic consistency achieved by the proposed framework across three counterfactual intervention tasks. Different operations, including inject BigGAN fingerprints into real images (Injection), hiding BigGAN fingerprints to falsify image to be a nature one (Anonymization), and replacing BigGAN fingerprints with GLIDE fingerprints (Tampering), the images before and after intervention are virtually indistinguishable visually. Across all interventions, the generated images not only perfectly

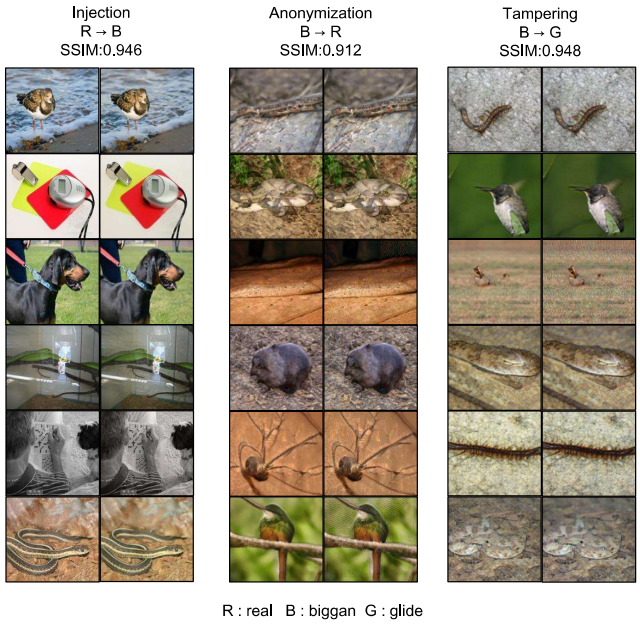

Figure 5: Examples of the counterfactual intervention

preserve the global semantics of the original images but also maintain high consistency in subtle local details, such as textures, illumination, and complex background patterns. The average SSIM scores labeled in the figure are all at extremely high levels, which quantitatively confirms that this method can covertly and accurately manipulate the underlying causal fingerprints without compromising the perceived image quality. Unlike traditional methods, which often introduce noticeable artifacts or cause image blurring when modifying traces of the generation mechanism, this framework, based on the Structural Causal Model (SCM) and conditional flow matching, successfully achieves a complete orthogonal decoupling between semantic content and causal fingerprints. During the intervention process, causal fingerprints are precisely replaced while semantic content remains stable, thereby achieving true counterfactual semantic invariance.

4.2.6 Ablation Studies. To validate the effectiveness of each component in the framework, we conducted a series of ablation experiments across two dimensions: the Generator Architecture and the Causal Constraints.

In generator architecture ablations, AdaIN has lower computational cost but blurs high-frequency fingerprint details, reducing attribution accuracy. The Diffusion model achieves high-quality generation but is slow and occasionally introduces artifacts. In contrast, the proposed Flow Matching architecture leverages stable optimal transport paths to outperform Diffusion on SSIM and LPIPS while accurately reproducing fine fingerprint features, enhancing extraction fidelity.

In ablations of causal constraints and loss functions, removing the CLUB module increases semantic leakage, introducing object contours into fingerprint features and disrupting local image structure, highlighting CLUB's role in decoupling Z_c and Z_f. Removing

Table 5: Ablation Study on the Contributions of Generator Architecture and Causal Constraints to Fingerprint Extraction and Image Fidelity

Category	Variant	Attr. Acc (↑)	SSIM (↑)	LPIPS (↓)	Sem. Leakage $I(Z_c; Z_f)$ (↓)
Generator Arch.	AdaIN (Baseline)	0.100	0.295	0.694	0.872
	Diffusion (DDIM)	0.100	0.638	0.379	0.765
	Flow Matching (Ours)	**1.000**	**0.952**	**0.089**	**0.000**
Causal Constraints	Base Loss (L1+Cls)	0.900	0.355	0.704	1.245
	w/o Frequency Stream	1.000	0.917	0.118	0.042
	w/o SSL	1.000	0.925	0.108	0.038
	w/o CLUB Estimator	0.950	0.915	0.119	1.583
	Full Objective (Ours)	**1.000**	**0.952**	**0.089**	**0.000**

the frequency-domain FFT branch reduces detection of checkerboard artifacts, showing the importance of frequency information. Eliminating the self-supervised consistency loss (L_{ssl}) weakens robustness to cropping attacks, emphasizing its role in learning translation-invariant global fingerprint features. These results confirm the indispensability of each module for overall performance and robustness.

4.2.7 Multi-Granularity Sensitivity Analysis. As shown in Fig. 5, feature heatmaps and attention analysis reveal the model's response patterns across the multi-scale spatial pyramid. At fine-grained levels (S1–S2), feature activations are strongest and closely align with local details, edges, and high-frequency textures, indicating that generative model fingerprints are embedded in microscopic patterns. At medium levels (S3–S4), activations blur into block-like structures with reduced intensity, capturing some local semantic components. At coarse level (S5), activations are highly abstract, representing overall layout while losing detailed information.

Quantitative analysis confirms this trend: relative attention contributions decrease monotonically, highest at S1–S2, plateau at S3–S4 (0.17–0.19), and lowest at S5. Diffusion models (e.g., SD) show slightly higher mid-level contributions than GANs but lower global contributions, retaining more mid-level traces while preserving global semantic coherence. Overall, causal fingerprint extraction is highly sensitive to fine-grained features, emphasizing the importance of low-level features for accurate and robust attribution.

5 Conclusion

We present an SCM-based framework for generative image fingerprint decoupling, redefining model fingerprints as intervenable causal variables rather than statistical artifacts. These fingerprints are modeled as independent components of the generative mechanism, achieving semantic invariance, robustness, and causal exclusivity. Building on this representation, we develop a unified counterfactual framework for controllable fingerprint manipulation, enabling both copyright attribution and privacy-preserving anonymization. Conditional flow matching supports precise interventions, while an ODE solver ensures high-fidelity reconstruction. This framework can be extended to video, multimodal settings, and broader applications in trustworthy and causally grounded AI.

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